

Question				Marking details	Marks available											
					AO1	AO2	AO3	Total	Maths	Prac						
4	(a)	(i)		compound (1) lower (1) electrical (1) liquid (1)	4			4								
		(ii)		Al ₂ O ₃		1		1								
		(iii)		<table border="1"><tr><td>carbon is more reactive than aluminium</td><td></td></tr><tr><td>iron is more reactive than aluminium</td><td></td></tr><tr><td>aluminium is more reactive than carbon</td><td>✓</td></tr></table>	carbon is more reactive than aluminium		iron is more reactive than aluminium		aluminium is more reactive than carbon	✓	1			1		
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iron is more reactive than aluminium																
aluminium is more reactive than carbon	✓															
	(b)			Indicative content electrical wiring - ductile – can be drawn into wires - good electrical conductor – current can pass through it saucepans - good thermal conductor – heat can pass through it - high melting point – can be heated to high temperatures - corrosion resistant – will not corrode - malleable – can be hammered into shape / shaped - non-toxic – safe to use for food	4		2	6								

Question				Marking details	Marks available						
					AO1	AO2	AO3	Total	Maths	Prac	
				water pipes • malleable – can be hammered into shape / shaped • corrosion resistant – will not corrode • non-toxic – safe for (drinking) water references to strength, durability, low/high density are not directly relevant to these uses 5-6 marks Description of two relevant properties linked to all three uses <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3-4 marks Description of one relevant property linked to all three uses <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Identification of one relevant property linked to one or two uses <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>							
				Question 4 total	9	1	2	12	0	0	